

MANGALORE UNIVERSITY

Department of Computer Science & Applications

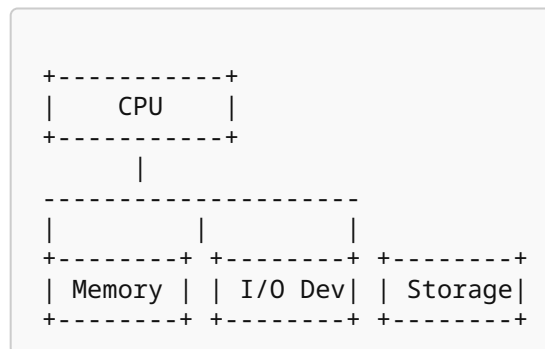
Course: Bachelor of Computer Applications (BCA) • Subject: Operating Systems

1. Explain Computer System Organization with Diagram.

INTRODUCTION

Computer System Organization refers to the arrangement of hardware components and their interaction. The operating system coordinates these components for efficient execution of programs.

DIAGRAM



MAIN POINTS

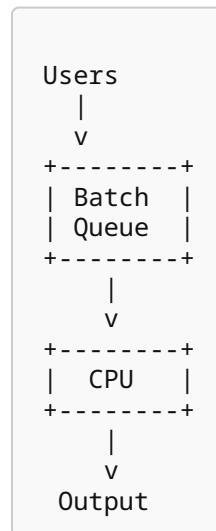
- 1. CPU (Central Processing Unit):** CPU is the brain of the computer system. It executes instructions and controls all system operations.
- 2. Main Memory:** Main memory stores programs and data currently being used. It provides fast access to the CPU.
- 3. Input Devices:** Input devices such as keyboard and mouse are used to enter data into the computer. They send information to the CPU for processing.
- 4. Output Devices:** Output devices such as monitor and printer display the processed results. They help users receive information from the system.
- 5. Secondary Storage:** Storage devices like hard disks permanently store programs and data. They retain information even when power is off.
- 6. System Bus:** The system bus connects CPU, memory, and I/O devices. It transfers data, instructions, and control signals.

2. Explain Batch Processing System.

INTRODUCTION

Batch Processing System is an operating system in which similar jobs are collected and executed as a group. User interaction is not required during execution.

DIAGRAM



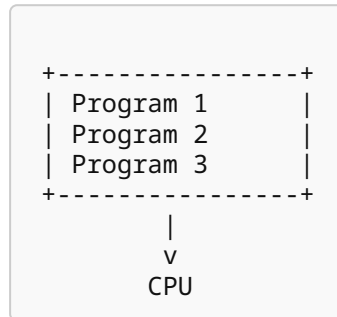
MAIN POINTS

- 1. Collection of Jobs:** Users submit jobs to the system operator. Similar jobs are grouped together into batches.
- 2. Sequential Execution:** Jobs are executed one after another automatically. No manual intervention is required during processing.
- 3. No User Interaction:** The user cannot interact with the job while it is running. Results are obtained only after execution is completed.
- 4. Better Resource Utilization:** The CPU processes a large number of jobs continuously. This improves system efficiency.
- 5. Suitable Applications:** Batch processing is useful for payroll systems and bill generation. These tasks involve repetitive processing.
- 6. Limitations:** It has long turnaround time and difficult error correction. Immediate response is not possible.

3. Write a Note on Multiprogrammed Systems.

INTRODUCTION

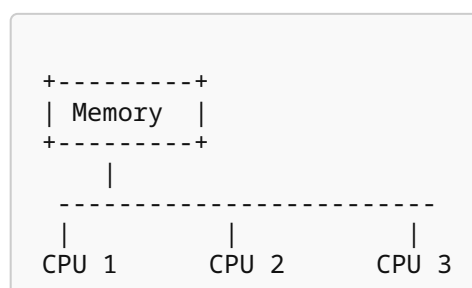
Multiprogramming is a technique in which multiple programs are kept in memory simultaneously. The CPU switches between programs to maximize utilization.

DIAGRAM**MAIN POINTS**

- 1. Multiple Programs in Memory:** Several programs are loaded into main memory at the same time. This allows efficient use of system resources.
- 2. CPU Scheduling:** The operating system selects one program for execution. The CPU switches whenever a program waits for I/O.
- 3. Improved CPU Utilization:** The CPU remains busy most of the time. Idle CPU time is greatly reduced.
- 4. Increased Throughput:** More jobs can be completed within a given period. Overall system performance improves.
- 5. Memory Management Requirement:** Proper memory allocation is necessary for multiple programs. The operating system manages memory efficiently.
- 6. Reduced Waiting Time:** Users experience faster job completion. System resources are utilized effectively.

4. Explain Multiprocessor System with Diagram.**INTRODUCTION**

A Multiprocessor System contains two or more processors sharing common memory and resources. These processors work together to improve performance.

DIAGRAM**MAIN POINTS**

- 1. Multiple CPUs:** The system contains more than one processor. Each processor can execute tasks independently.
- 2. Shared Memory:** All processors share the same main memory. Data can be accessed by any processor.

- 3. Parallel Processing:** Multiple tasks can be executed simultaneously. This increases system speed and efficiency.
- 4. Increased Reliability:** Failure of one processor may not stop the entire system. Other processors continue operation.
- 5. Resource Sharing:** Processors share memory and I/O devices. This reduces hardware cost.
- 6. Types of Multiprocessor Systems:** Multiprocessor systems are classified as SMP and AMP. Both improve overall system performance.

5(a). Real Time System

INTRODUCTION

A Real Time System is an operating system that provides responses within a specified time limit. Timely execution is more important than high throughput.

MAIN POINTS

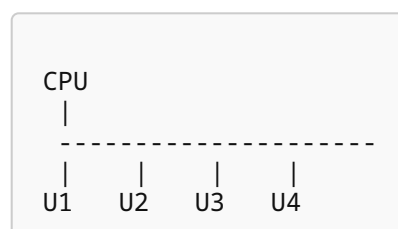
- 1. Immediate Response:** The system responds to events without significant delay. Deadlines must be satisfied.
- 2. Deterministic Behaviour:** Execution time is predictable and fixed. This ensures reliable operation.
- 3. High Reliability:** The system is designed for critical applications. Failure can lead to serious consequences.
- 4. Hard Real Time System:** Missing a deadline is unacceptable in hard real-time systems. Examples include aircraft control systems.
- 5. Soft Real Time System:** Missing a deadline slightly affects performance. Multimedia systems are common examples.
- 6. Applications:** Real-time systems are used in robotics, medical equipment, and industrial control systems. They require fast and accurate responses.

5(b). Time Sharing System

INTRODUCTION

Time Sharing System allows multiple users to access a computer simultaneously. CPU time is divided among users in small time slices.

DIAGRAM



MAIN POINTS

1. **Time Slicing:** CPU time is divided into small units called time slices. Each user receives a share of CPU time.
2. **Multuser Environment:** Many users can work on the same system simultaneously. Each user feels they have exclusive access.
3. **Quick Response Time:** Users receive immediate responses to commands. Interactive computing becomes possible.
4. **Efficient Resource Utilization:** CPU and memory are shared among users. System resources are used effectively.
5. **Context Switching:** The operating system rapidly switches between users. This creates the illusion of parallel execution.
6. **Applications:** Time-sharing systems are commonly used in servers and educational institutions. They support multiple users efficiently.

6. Explain the Functions of Following Components of the Operating System

a) Process Management

INTRODUCTION

Process Management is responsible for creating, scheduling, and terminating processes. It ensures efficient utilization of the CPU.

MAIN POINTS

1. **Process Creation and Deletion:** The operating system creates new processes when programs are executed. It also removes processes after completion.
2. **CPU Scheduling:** The OS selects which process should get CPU time. This improves system performance and responsiveness.
3. **Process Synchronization:** Synchronization coordinates multiple processes accessing shared resources. It prevents data inconsistency.
4. **Deadlock Handling:** The OS detects and prevents deadlocks among processes. This ensures smooth execution.
5. **Inter-Process Communication:** Processes communicate and exchange data through IPC mechanisms. This helps in coordinated execution.

b) Memory Management

INTRODUCTION

Memory Management controls the allocation and utilization of main memory. It ensures efficient use of available memory resources.

MAIN POINTS

- 1. Memory Allocation:** The OS allocates memory space to processes as needed. This allows programs to execute properly.
- 2. Memory Deallocation:** Memory occupied by completed processes is released. The freed memory becomes available for other processes.
- 3. Memory Tracking:** The OS keeps track of used and free memory spaces. This prevents memory wastage.
- 4. Virtual Memory Management:** Virtual memory extends available memory using secondary storage. Large programs can execute efficiently.
- 5. Protection and Sharing:** The OS protects one process from accessing another's memory. It also allows controlled memory sharing.

c) File Management

INTRODUCTION

File Management controls the creation, storage, and access of files. It organizes data efficiently on storage devices.

MAIN POINTS

- 1. File Creation and Deletion:** The OS creates new files and removes unwanted files. This helps manage user data.
- 2. File Organization:** Files are arranged systematically within directories. This makes retrieval easier.
- 3. Access Control:** The OS controls permissions for reading and writing files. Unauthorized access is prevented.
- 4. File Storage Management:** Storage space is allocated for files on disks. Efficient allocation improves performance.
- 5. Backup and Recovery:** The OS provides mechanisms for data recovery. Important files can be restored if lost.

d) Secondary Storage Management

INTRODUCTION

Secondary Storage Management handles permanent storage devices such as hard disks and SSDs. It manages free and occupied storage space.

MAIN POINTS

- 1. Space Allocation:** Disk space is allocated to files and directories. Efficient allocation improves storage utilization.
- 2. Free Space Management:** The OS keeps track of free disk blocks. These blocks are assigned when new files are created.

- 3. Disk Scheduling:** Disk requests are scheduled to reduce access time. This improves system performance.
- 4. Storage Organization:** Data is organized systematically on storage devices. Retrieval becomes faster and easier.
- 5. Reliability of Data:** The OS ensures data integrity and consistency. This reduces the risk of data loss.

e) Networking Management

INTRODUCTION

Networking Management enables communication between computers connected through a network. It manages data transmission and network resources.

MAIN POINTS

- 1. Communication Management:** The OS facilitates communication between networked systems. Data is transferred efficiently.
- 2. Resource Sharing:** Users can share files, printers, and other resources. This reduces duplication of resources.
- 3. Security Management:** The OS protects network resources from unauthorized access. Security measures safeguard data.
- 4. Protocol Support:** The OS supports networking protocols such as TCP/IP. These protocols ensure reliable communication.
- 5. Remote Access:** Users can access resources from remote locations. This increases flexibility and productivity.

f) I/O System Management

INTRODUCTION

I/O System Management controls communication between the computer and peripheral devices. It ensures efficient handling of input and output operations.

MAIN POINTS

- 1. Device Management:** The OS manages devices such as keyboards, printers, and monitors. Proper coordination is maintained.
- 2. Device Drivers:** Drivers act as interfaces between hardware and software. They enable smooth device operation.
- 3. Buffering:** Temporary storage areas are used during data transfer. This improves I/O efficiency.
- 4. Interrupt Handling:** The OS responds to hardware interrupts generated by devices. This ensures timely processing.
- 5. I/O Scheduling:** Input and output requests are scheduled efficiently. This minimizes waiting time.

7. Explain Any Five Operating System Services

INTRODUCTION

Operating system services provide facilities for users and programs. They make computer systems convenient and efficient to use.

MAIN POINTS

- 1. Program Execution:** The OS loads programs into memory and executes them. It also handles program termination.
- 2. I/O Operations:** The OS manages communication with I/O devices. Users can perform input and output easily.
- 3. File System Manipulation:** Files can be created, modified, deleted, and accessed. The OS maintains file organization.
- 4. Communication:** Processes can exchange information through communication mechanisms. This supports multitasking.
- 5. Error Detection:** The OS detects hardware and software errors. Appropriate corrective actions are taken.

8. Explain Any Five File Operations

INTRODUCTION

File operations are actions performed on files by the operating system. They help users manage and access data efficiently.

MAIN POINTS

- 1. Create:** A new file is created and allocated storage space. The file becomes available for use.
- 2. Open:** Opening a file establishes access to its contents. The OS loads file information into memory.
- 3. Read:** Data is retrieved from a file and transferred to memory. Users can view stored information.
- 4. Write:** New data is stored in a file. Existing data may also be modified.
- 5. Delete:** The file is removed from storage. The occupied space is released for future use.

9. Explain Various File Attributes

INTRODUCTION

File attributes are characteristics that describe a file. They help the operating system manage files effectively.

MAIN POINTS

- 1. File Name:** The file name uniquely identifies a file. Users use it to access the file.
- 2. File Type:** The type indicates the nature of the file. Examples include text, executable, and image files.
- 3. File Size:** It specifies the amount of storage occupied by the file. Size is usually measured in bytes.

- 4. Location:** The location indicates where the file is stored on disk. The OS uses this information for access.
- 5. Protection Information:** Protection attributes define access permissions. They control who can read, write, or execute the file.
- 6. Date and Time Information:** The OS stores creation, modification, and access dates. These attributes help track file activity.

10. Explain Any Five Common File Types, Extensions and Their Functions

INTRODUCTION

Different file types are used to store different kinds of data. Each file type is identified by a specific extension.

MAIN POINTS

File Type	Extension	Function
Text File	<i>.txt</i>	Stores plain text information without formatting. It is commonly used for notes and documents.
Document File	<i>.doc, .docx</i>	Stores formatted text documents. It is widely used in word processing applications.
Image File	<i>.jpg, .png</i>	Stores digital images and graphics. These files are used for photographs and pictures.
Audio File	<i>.mp3</i>	Stores sound and music data. It is commonly used for audio playback.
Executable File	<i>.exe</i>	Contains program instructions that can be executed by the computer. It is used to run software applications.

11. Explain Various Access Methods in Files

INTRODUCTION

File access methods determine how data is read from or written to a file. Different access methods are used depending on the application requirements.

MAIN POINTS

- 1. Sequential Access:** Records are accessed one after another in sequence. To reach a specific record, all previous records must be read.
- 2. Direct (Random) Access:** Any record can be accessed directly without reading previous records. This method provides faster data retrieval.

3. **Indexed Access:** An index is maintained containing pointers to file records. The index helps locate records quickly.
4. **Indexed Sequential Access:** This method combines sequential and indexed access techniques. Records can be accessed both sequentially and directly.
5. **Efficiency of Access Methods:** Different methods are chosen based on application needs. Proper selection improves file performance and storage utilization.

12. Explain Any Five Operations that can be Performed on Directory

INTRODUCTION

A directory is a collection of files and subdirectories. The operating system provides several operations to manage directories efficiently.

MAIN POINTS

1. **Create Directory:** A new directory can be created to organize files. It helps maintain a structured file system.
2. **Delete Directory:** Unwanted directories can be removed from the system. This releases storage space.
3. **Search Directory:** Users can search for files within a directory. The operating system locates the required file quickly.
4. **Rename Directory:** The name of an existing directory can be changed. This improves organization and identification.
5. **List Directory Contents:** The operating system displays files and subdirectories present in a directory. This helps users manage stored data.

13. Explain Different Types of Access Control for File Operations

INTRODUCTION

Access control determines who can access a file and what operations can be performed on it. It ensures file security and protection.

MAIN POINTS

1. **Read Access:** Read permission allows a user to view the contents of a file. No modification is permitted.
2. **Write Access:** Write permission allows users to modify file contents. New information can be added or existing data can be changed.
3. **Execute Access:** Execute permission allows a file to run as a program. It is mainly used for executable files.
4. **Append Access:** Append permission allows users to add data at the end of a file. Existing contents cannot be modified.

- 5. Delete Access:** Delete permission allows users to remove files from storage. Unauthorized deletion is prevented through access control.

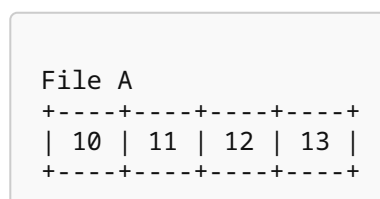
14. Explain Any Two Allocation Methods

(a) Contiguous Allocation

INTRODUCTION

Contiguous allocation stores all file blocks in consecutive disk locations. It is one of the simplest file allocation methods.

DIAGRAM



MAIN POINTS

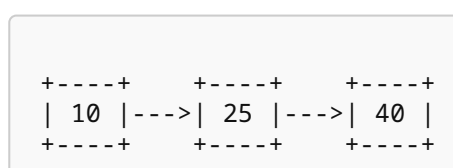
- 1. Consecutive Storage:** All blocks of a file are stored next to each other. This simplifies file management.
- 2. Fast Access:** Files can be accessed quickly because blocks are stored continuously. Both sequential and direct access are efficient.
- 3. Simple Implementation:** The operating system only needs the starting block and file length. File retrieval becomes easier.
- 4. External Fragmentation:** Free space may become scattered over time. Large files may not find sufficient contiguous space.
- 5. Better Performance:** Disk head movement is reduced significantly. This improves overall file access speed.

(b) Linked Allocation

INTRODUCTION

Linked allocation stores file blocks at different disk locations. Each block contains a pointer to the next block.

DIAGRAM



MAIN POINTS

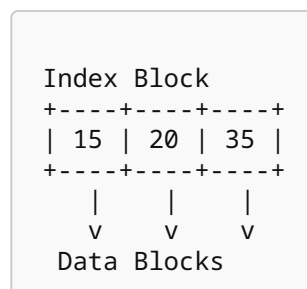
- 1. Non-Contiguous Storage:** File blocks can be stored anywhere on the disk. Contiguous free space is not required.
- 2. Linked Structure:** Each block contains the address of the next block. This creates a linked list of blocks.
- 3. No External Fragmentation:** Since blocks can be stored anywhere, external fragmentation is avoided. Storage utilization improves.
- 4. Sequential Access Support:** Linked allocation is suitable for sequential file access. Files can be read block by block.
- 5. Pointer Overhead:** Extra space is required for storing pointers. This slightly reduces usable storage space.

(c) Indexed Allocation

INTRODUCTION

Indexed allocation uses a separate index block to store addresses of all file blocks. This supports direct file access.

DIAGRAM



MAIN POINTS

- 1. Index Block Usage:** Each file has a dedicated index block. It stores addresses of all file blocks.
- 2. Direct Access:** Any block can be accessed directly through the index. This improves retrieval speed.
- 3. No External Fragmentation:** Blocks can be stored anywhere on the disk. Continuous storage space is not required.
- 4. Efficient File Management:** The operating system can locate file blocks quickly. File access becomes more efficient.
- 5. Additional Storage Requirement:** Extra space is needed for index blocks. Large files may require multiple index blocks.

15. Explain Any Two Free-Space Management Methods

(a) Bit Vector Method

INTRODUCTION

The Bit Vector method uses bits to represent free and occupied disk blocks. Each block is associated with one bit.

MAIN POINTS

- 1. Representation of Blocks:** A bit value of 0 may indicate a free block. A bit value of 1 indicates an allocated block.
- 2. Easy Identification:** Free blocks can be identified quickly by examining the bit map. Allocation becomes efficient.
- 3. Compact Storage:** Only one bit is required for each disk block. This reduces storage overhead.
- 4. Efficient Searching:** The operating system can search for free blocks rapidly. Contiguous free blocks can also be found easily.
- 5. Widely Used Method:** Many modern operating systems use bit vectors. It provides efficient free-space management.

(b) Linked List Method

INTRODUCTION

In the Linked List method, all free disk blocks are linked together. Each free block contains a pointer to the next free block.

MAIN POINTS

- 1. Linked Free Blocks:** All available blocks are connected through pointers. This forms a linked list.
- 2. Free Space Tracking:** The operating system maintains a pointer to the first free block. New allocations use blocks from this list.
- 3. Dynamic Allocation:** Free blocks can be allocated whenever required. Storage management becomes flexible.
- 4. Reduced Memory Requirement:** No separate bitmap is needed. Only pointers are maintained.
- 5. Slower Searching:** Finding large groups of free blocks may take more time. The list must be traversed sequentially.

OTHER FREE-SPACE MANAGEMENT METHODS

Grouping Method

Counting Method

(Note: Write any two methods in the exam as asked.)

End of Examination Syllabus Handout